Analysis Phase – Polyline Editor

System Perspective:

The Polyline Editor is an interactive system where users perform tasks through direct manipulation (mouse) and mode-based commands (keyboard). The interaction is user-driven, system responds to user inputs and provides immediate visual feedback. The focus is on efficient task execution and low cognitive load.

Core Interaction Objects

The interaction is centered around the following entities:

- **Polyline**: structured object representing connected line segments
- **Vertex (Point)**: fundamental unit of interaction
- **Line Segment**: visual connection between vertices
- **Canvas**: primary interaction space
- **Cursor**: pointing and selection tool
- **Mode State**: defines current system behavior
- **Input Events**: mouse clicks, keyboard presses

User Analysis

Skill Levels

Novice Users (Beginners):

- Limited system knowledge
- Rely on **recognition over recall**
- Require clear instructions, visual guidance, and continuous feedback

Intermediate Users:

- Partial familiarity with system
- Use a mix of **recall and recognition**
- Benefit from both visual cues and shortcuts

Expert Users:

- Highly experienced
- Prefer **efficiency and low interaction cost**
- Rely on keyboard shortcuts, muscle memory, and direct manipulation

User Task Analysis

Create Polyline

- User activates begin mode ('b')
- Performs repeated mouse clicks
- System generates vertices and connects them

Move Vertex

- User activates move mode ('m')
- Selects nearest vertex
- Drags or repositions it

Delete Vertex

- User activates delete mode ('d')
- System identifies nearest vertex
- Removes it and reconnects adjacent segments

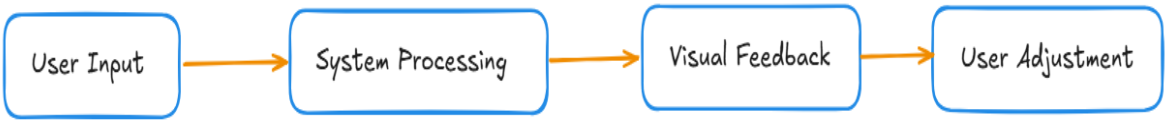
Refresh Display

- User presses 'r'
- System redraws all objects

Exit Interaction

- User presses ‘q’

Interaction Flow Model



This reflects Norman’s execution–evaluation cycle:

- execution (user action)
- evaluation (system feedback)

Error and Exception Analysis

Possible User Errors:

- selecting wrong vertex
- accidental deletion
- confusion between modes
- clicking outside valid range

System Response Should:

- minimize errors (error prevention)
- tolerate mistakes (error recovery)
- provide clear feedback

Data Representation and Structural Model

- Polylines are stored as ordered data structures (we used array)
- Each polyline consists of sequential vertices

```
const canvas = document.getElementById("canvas");
const ctx = canvas.getContext("2d");

let polylines = [];
```

Usability and Interaction Considerations

- **Visibility:** system state (mode) must be clear
- **Feedback:** immediate visual updates
- **Consistency:** uniform interaction across tasks
- **User Control:** ability to modify and correct actions
- **Learnability:** simple commands reduce learning effort
- **Efficiency:** minimal steps for common tasks

Interaction Environment Analysis

Touchpad Users

- Low precision → higher error rate
- Need larger hit areas and hover feedback

Mouse Users

- Moderate precision
- Benefit from visual feedback and consistency

Keyboard-Oriented Users

- Prefer shortcuts

- Require fast response and minimal latency

Small Screen Environments

- Limited space → risk of clutter
- Requires minimalist and structured layout